MUHSINA KOTTARATH

AI Engineer

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Profile

Data professional with 4+ years across machine learning, LLM applications and production data engineering, plus an MSc in Data Science (Distinction) with a computer vision dissertation. Takes models beyond the notebook: deep learning for image classification, RAG systems built from scratch. Strong Python and SQL, with hands-on PySpark, Databricks, Docker and basic cloud experience across AWS and Azure.

Technical Skills

Generative AI: LLMs, Retrieval-Augmented Generation (RAG), Haystack, prompt engineering, embeddings, vector search, AI agent deployment, MCP

Machine learning: Deep learning, CNNs, computer vision, image classification, feature extraction, classification and predictive modelling, model evaluation, time series, anomaly detection

Languages and Python stack: Python, SQL, pandas, NumPy, matplotlib, PySpark

Data engineering: Apache Spark, Databricks, Delta Lake, dbt, ETL/ELT pipelines, feature engineering, data modelling, data quality and validation

Cloud and tooling: AWS, Azure, Docker, Git, CI/CD (YAML), Linux, PostgreSQL

Experience

Artificial Intelligence Engineer (Part-time contract) | Ticket Buddy | Apr 2026 - Sep 2026 | Remote, UK

- Supported RAG pipelines in Haystack for an AI-powered customer support product, so the AI answers customer queries from a structured knowledge base.
- Built automated Python pipelines that process documents, web content, emails and third-party data into that knowledge base.
- Worked with backend and AI engineers on integration, and ran validation and quality checks on the data feeding the AI.

Data Analyst | Deloitte | Aug 2025 - Feb 2026 | Newcastle upon Tyne, UK

- Processed and validated large-scale financial and audit datasets (1M+ records) using PySpark on Databricks, building structured data quality checks for accuracy and consistency across audit outputs.
- Mapped and standardised raw audit data into a cloud platform schema (Omnia), resolving inconsistencies in entity structures, date formats and column mappings across audit teams in Canada, Germany and the Netherlands.
- Escalated and resolved a reconciliation risk by showing stakeholders the data integrity impact of removing unmapped ledger accounts, securing the correct source data instead of a workaround.
- Managed 5 concurrent data tickets internationally at 99% quality accuracy during peak financial year-end periods.

Prompt Engineer | Outlier (AI Startup) Part-time | Sep 2024 - Apr 2025 | Newcastle upon Tyne, UK

- Evaluated and compared multiple LLM responses, validating outputs against reliable sources and logical checks to identify reasoning and factual failure modes.
- Improved prompt quality through iterative testing and structured feedback, contributing to more consistent model outputs.

Data Science Associate (Intern) | Martell Real Estate | Sep 2024 - Jan 2025 | Newcastle upon Tyne, UK

- Led a team of 5 to design and build a full-stack analytics platform (Django, PostgreSQL, HTML/CSS) analysing UK government company insolvency data from 2021 to 2023 for an external client, with AI tools integrated for automated analysis.
- Built an automated ETL process in Python to read, clean, deduplicate and load source data into PostgreSQL, with checks preventing duplicate or overwritten records.
- Presented findings to internal and external stakeholders throughout delivery, cutting development time by 30% and delivering 100% on time.

Data Engineer | Tranzmeo (Startup) | Aug 2020 - Mar 2022 | Kochi, India

- Developed real-time Kafka streaming pipelines ingesting time-series sensor data, supporting anomaly detection on oil and gas pipelines for a national petroleum client.
- Diagnosed missing and inconsistent values degrading model inputs; introduced pre-collection validation with completeness alerts and applied median imputation where loss was under 30%, improving the reliability of the anomaly detection feed.
- Built and maintained ETL pipelines in Python and PostgreSQL for sensor data across 60+ pipeline monitoring stations, improving processing efficiency by 20%.
- Led migration of the on-premise platform from Hadoop to Apache Spark/PySpark, independently learning and deploying Spark, Kafka and Docker across all pipelines within 3-4 weeks.
- Automated daily data validation and infrastructure health checks in Python, saving 40 minutes per day, and implemented CI/CD (YAML) to improve deployment reliability.

Data Research Associate | Fullcontact | Dec 2019 - Jun 2020 | Kochi, India

- Validated 600+ algorithm-generated data entries daily using logical checks and QC processes, maintaining over 90% data accuracy.

Machine Learning and AI Projects

RAG Pipeline with Open-Source Tools | Python, LLMs, Vector Search

- Built a retrieval-augmented generation chatbot from scratch, integrating a vector database with a local lightweight language model to ground responses in a custom knowledge base.
- Implemented the full pipeline (ingestion, embedding, retrieval and prompt design) and optimised it to run efficiently on limited hardware.

Multi-Feature Deepfake Detection System (MSc Dissertation) | Python, CNN, Computer Vision

- Designed an image classification system combining a MesoNet CNN with Local Binary Pattern texture feature extraction to distinguish real from manipulated images.
- Evaluated each technique independently before combining them; the multi-feature model outperformed either approach alone.

UK Crime Analytics | PySpark, Azure, Time Series and Regression

- Built a scalable analytics pipeline processing 65M+ records, surfacing trends in violent crime and correlations between drug and firearm incidents.

Education

MSc Data Science with Advanced Practice, Distinction | Northumbria University, Newcastle upon Tyne | 2022 - 2024

Machine Learning, Deep Learning, Data Analysis, Data Visualisation. Dissertation in computer vision (deepfake image detection).

BSc Computer Science and Engineering | University of Calicut | 2014 - 2018

Certifications

Local AI: Build a RAG Model from Scratch with Open-Source Tools, Big Data Analytics and Machine Learning , Python for Data Science and Machine Learning Bootcamp, Data Engineering with dbt ,Data Engineering Foundations, Career Essentials in Generative AI, Web Scraping and API Fundamentals in Python